# PRACTICAL 8

## Aim :

Improve React frontend performance using route-based lazy loading and code splitting. Load page components only when their routes are visited, provide a Suspense fallback while a chunk loads, and compare the build output before and after optimization.

## Code :

**frontend/src/App.jsx :-**

```
import { lazy, Suspense, useEffect, useState } from 'react';
import { Routes, Route } from 'react-router-dom';

const Home = lazy(() => import('./pages/Home'));
const Projects = lazy(() => import('./pages/Projects'));
const Contact = lazy(() => import('./pages/Contact'));
const NotFound = lazy(() => import('./pages/NotFound'));

function App() {
  const skillList = ['React', 'Node.js', 'Express', 'MongoDB', 'JavaScript'];

  return (
        {/* Header and navigation remain available immediately. */}

          Loading page...

}>

                    } />

          } />

          } />

          } />

                      );

}

export default App;
```

## Implementation :

**frontend/src/App.css :-**

```
.route-loading {
  min-height: 180px;
```

```
  display: grid;
  place-items: center;
  font-size: 1.1rem;
}
```

**Build commands :-**

cd frontend

npm run build

npm run dev

**Code-splitting behavior :-**

Initial bundle: shared application code

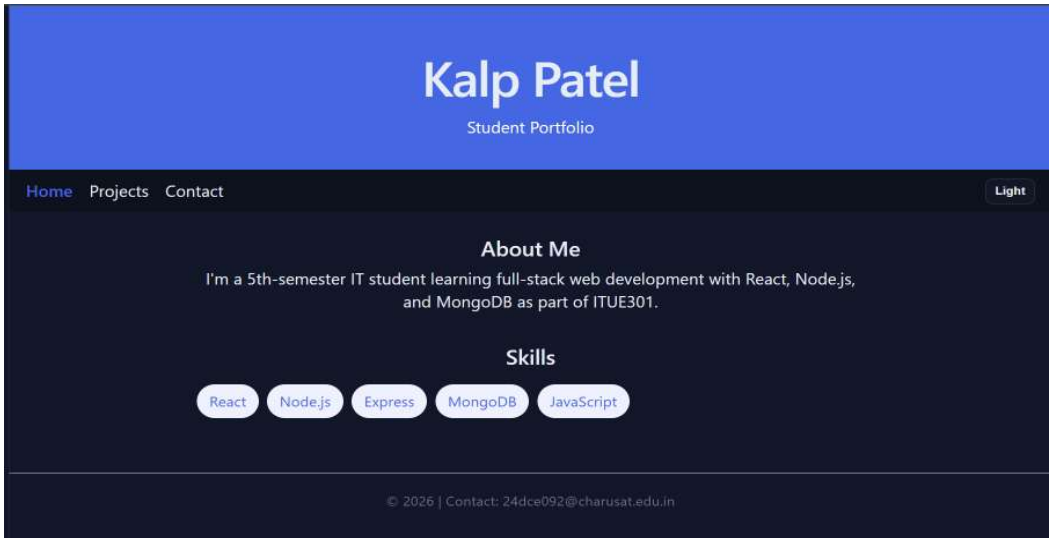
Home chunk: loaded when / is visited

Projects chunk: loaded when /projects is visited

Contact chunk: loaded when /contact is visited

NotFound chunk: loaded when an unknown route is visited

# Output :




```
const Home = lazy(() => import('./pages/Home'));
const Projects = lazy(() => import('./pages/Projects'));
const Contact = lazy(() => import('./pages/Contact'));
const NotFound = lazy(() => import('./pages/NotFound'));

function App() {
  const skillList = ['React', 'Node.js', 'Express', 'MongoDB', 'JavaScript'];
  const [theme, setTheme] = useState(() => {···
  });

  useEffect(() => {···
  }, [theme]);

  function toggleTheme() {···
  }

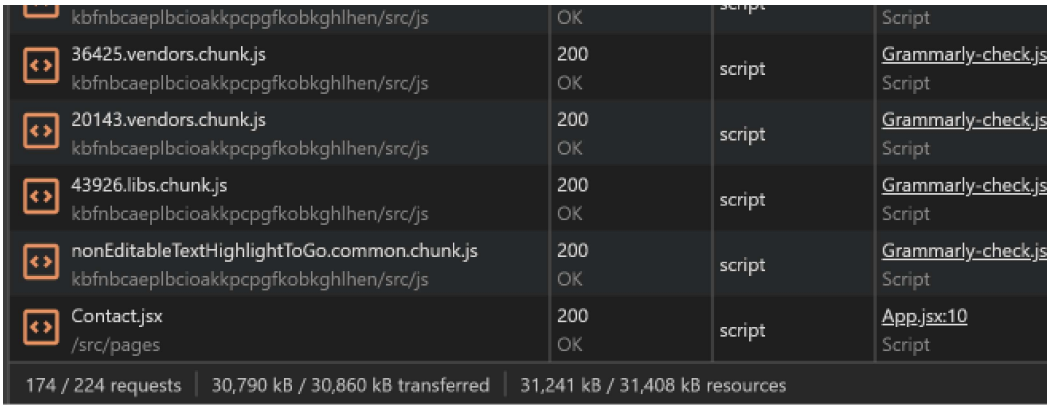
  return (
    <div className={`app-root ${theme}`}>
      <Header name="Kalp Patel" themeColor="#2563eb" />
      <NavBar onToggleTheme={toggleTheme} theme={theme} />
      <main>
        <Suspense fallback={<div className="route-loading" role="status">Loading page...</div>}>
          <Routes>
```

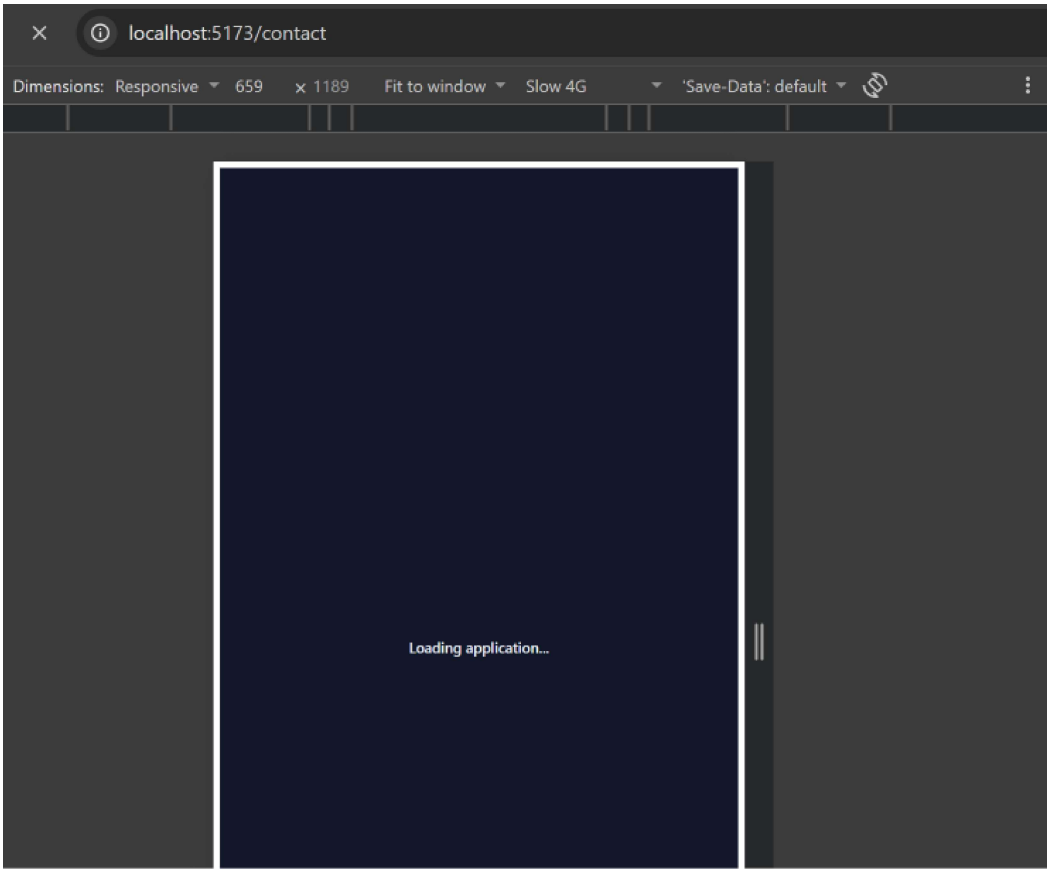
```
PS C:\Users\patel\OneDrive\Documents\SEM 5\AWD> cd frontend
>> npm run build

> frontend@0.0.0 build
> vite build

vite v8.2.1 building client environment for production...
✓ 39 modules transformed.
computing gzip size...
dist/index.html                  0.45 kB | gzip:  0.29 kB
dist/assets/index-FMXr24SE.css   3.19 kB | gzip:  1.28 kB
dist/assets/NotFound-HqA9tUuQ.js 0.24 kB | gzip:  0.20 kB
dist/assets/Home-BbtsFQjA.js     0.63 kB | gzip:  0.33 kB
dist/assets/Contact-DiBflZxT.js  0.71 kB | gzip:  0.41 kB
```

# Output :






# Result :

Route-based code splitting was implemented successfully with React.lazy and Suspense. The build generated separate chunks for the route pages, reducing the initial JavaScript bundle and loading page code only when needed. A loading fallback is displayed while a lazy route is being downloaded.